AR Safety Mirror:
Advancing Cyclists' Rear Hazard Awareness with AR Camera

Martin Karastoyanov, Jueon Park, Wonseok Jang
Team 1

Date 23 September, 2025

Motivation

How do cyclists monitor their rear surroundings for potential hazards?

Need for Rear Awareness

Traditional solutions often fall short in providing comprehensive and safe rearward visibility

Traditional Mirrors

Limited field of view, prone to vibrations.

Head Turns

Loss of forward vision, unstable riding.

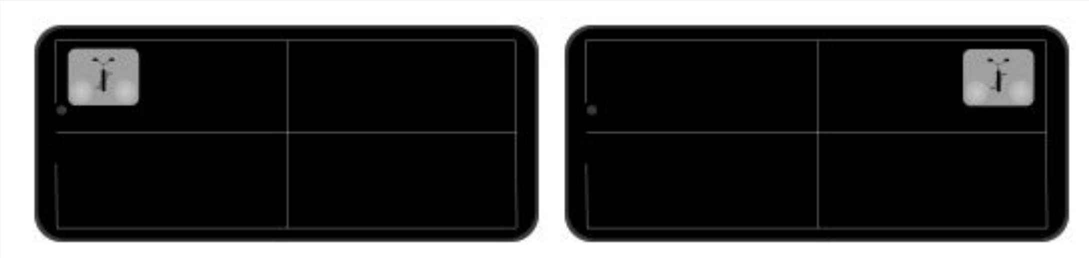
Missing Alerts of Others

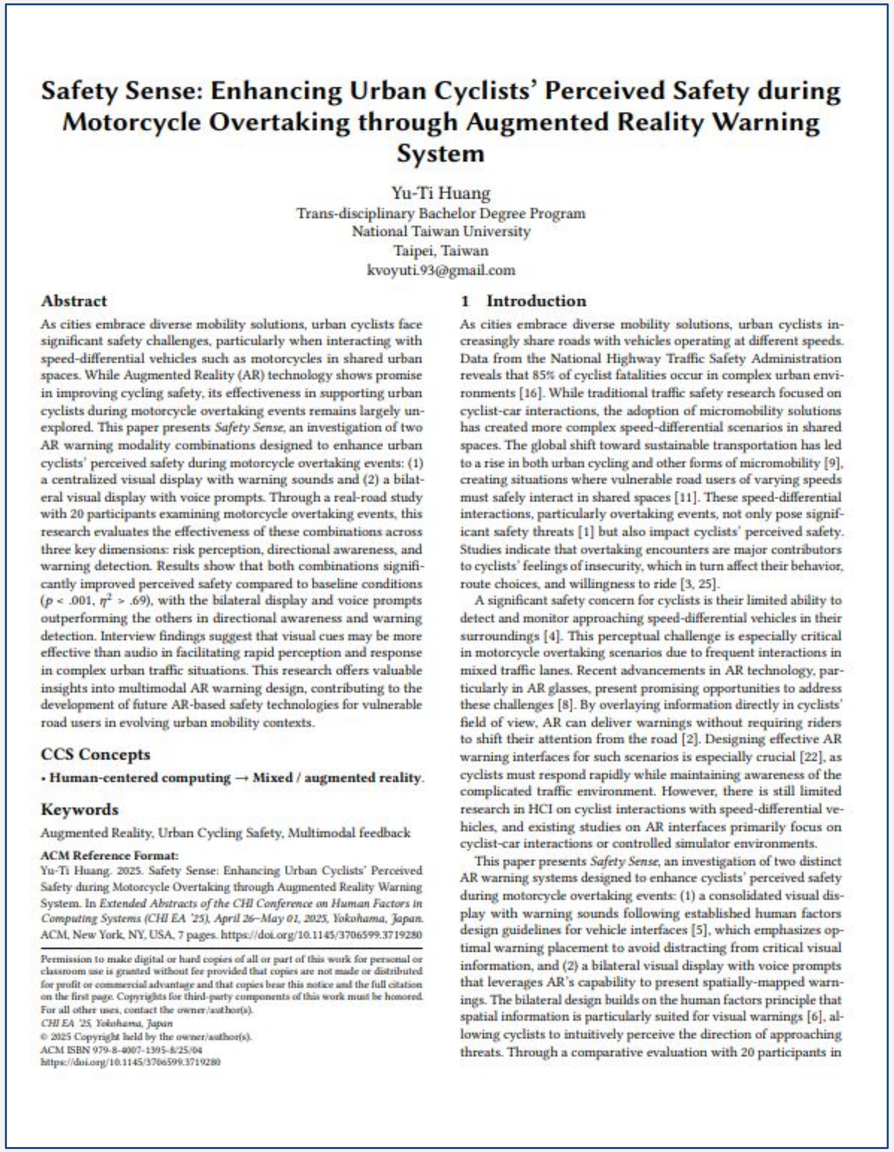
Can miss signals due to noisy outdoor environment

Motivation: Existing Paper

(1) Detection: ultrasonic sensor to detect overtaking motorcycles on the left/right side

(2) Alerts: visual display + warning sounds on AR glasses





Safety Sense: Enhancing Urban Cyclists' Perceived Safety during Motorcycle Overtaking through Augmented Reality Warning System

Yu-Ti Huang
Trans-disciplinary Bachelor Degree Program
National Taiwan University
Taipei, Taiwan
kvoyuti.93@gmail.com

Abstract

As cities embrace diverse mobility solutions, urban cyclists face significant safety challenges, particularly when interacting with speed-differential vehicles such as motorcycles in shared urban spaces. While Augmented Reality (AR) technology shows promise in improving cycling safety, its effectiveness in supporting urban cyclists during motorcycle overtaking events remains largely unexplored. This paper presents *Safety Sense*, an investigation of two AR warning modality combinations designed to enhance urban cyclists' perceived safety during motorcycle overtaking events: (1) a centralized visual display with warning sounds and (2) a bilateral visual display with voice prompts. Through a real-road study with 20 participants examining motorcycle overtaking events, this research evaluates the effectiveness of these combinations across three key dimensions: risk perception, directional awareness, and warning detection. Results show that both combinations significantly improved perceived safety compared to baseline conditions ($p < .001$, $\eta^2 > .69$), with the bilateral display and voice prompts outperforming the others in directional awareness and warning detection. Interview findings suggest that visual cues may be more effective than audio in facilitating rapid perception and response in complex urban traffic situations. This research offers valuable insights into multimodal AR warning design, contributing to the development of future AR-based safety technologies for vulnerable road users in evolving urban mobility contexts.

CCS Concepts

- **Human-centered computing → Mixed / augmented reality**.

Keywords

Augmented Reality, Urban Cycling Safety, Multimodal feedback

ACM Reference Format:
Yu-Ti Huang. 2025. Safety Sense: Enhancing Urban Cyclists' Perceived Safety during Motorcycle Overtaking through Augmented Reality Warning System. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25), April 26–May 01, 2025, Yokohama, Japan.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3706599.3719280

ACM ISBN 979-8-4007-1395-8/25/04
https://doi.org/10.1145/3706599.3719280

1 Introduction

As cities embrace diverse mobility solutions, urban cyclists increasingly share roads with vehicles operating at different speeds. Data from the National Highway Traffic Safety Administration reveals that 85% of cyclist fatalities occur in complex urban environments [16]. While traditional traffic safety research focused on cyclist-car interactions, the adoption of micromobility solutions has created more complex speed-differential scenarios in shared spaces. The global shift toward sustainable transportation has led to a rise in both urban cycling and other forms of micromobility [9], creating situations where vulnerable road users of varying speeds must safely interact in shared spaces [11]. These speed-differential interactions, particularly overtaking events, not only pose significant safety threats [1] but also impact cyclists' perceived safety. Studies indicate that overtaking encounters are major contributors to cyclists' feelings of insecurity, which in turn affect their behavior, route choices, and willingness to ride [3, 25].

A significant safety concern for cyclists is their limited ability to detect and monitor approaching speed-differential vehicles in their surroundings [4]. This perceptual challenge is especially critical in motorcycle overtaking scenarios due to frequent interactions in mixed traffic lanes. Recent advancements in AR technology, particularly in AR glasses, present promising opportunities to address these challenges [8]. By overlaying information directly in cyclists' field of view, AR can deliver warnings without requiring riders to shift their attention from the road [2]. Designing effective AR warning interfaces for such scenarios is especially crucial [22], as cyclists must respond rapidly while maintaining awareness of the complicated traffic environment. However, there is still limited research in HCI on cyclist interactions with speed-differential vehicles, and existing studies on AR interfaces primarily focus on cyclist-car interactions or controlled simulator environments.

This paper presents *Safety Sense*, an investigation of two distinct AR warning systems designed to enhance cyclists' perceived safety during motorcycle overtaking events: (1) a consolidated visual display with warning sounds following established human factors design guidelines for vehicle interfaces [5], which emphasizes optimal warning placement to avoid distracting from critical visual information, and (2) a bilateral visual display with voice prompts that leverages AR's capability to present spatially-mapped warnings. The bilateral design builds on the human factors principle that spatial information is particularly suited for visual warnings [6], allowing cyclists to intuitively perceive the direction of approaching threats. Through a comparative evaluation with 20 participants in

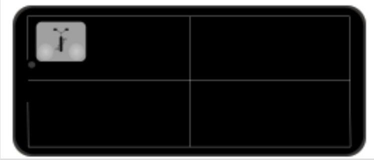
Limitations

Limited Field of View

Ultrasonic sensors has a **narrow detection cone**, only a partial view of the cyclist's rear surroundings. This can lead to blind spots and missed threats.

Inability to Differentiate

The current UI only shows that something is nearby, so riders still **need to look back** to see what it is



Short Detection Range

With a maximum range of only 4m, users have very **little time to react** to approaching hazards.

HC-SR04 Ultrasonic sensors with 4-meter max range

Objectives

Enhanced Hazard Detection

Camera for a significantly wider field of view and visual representation of the hazards, not just their presence.

Informative Mirror Interface

Familiar back-mirror interface with comprehensive information of potential hazards
=> direction, proximity

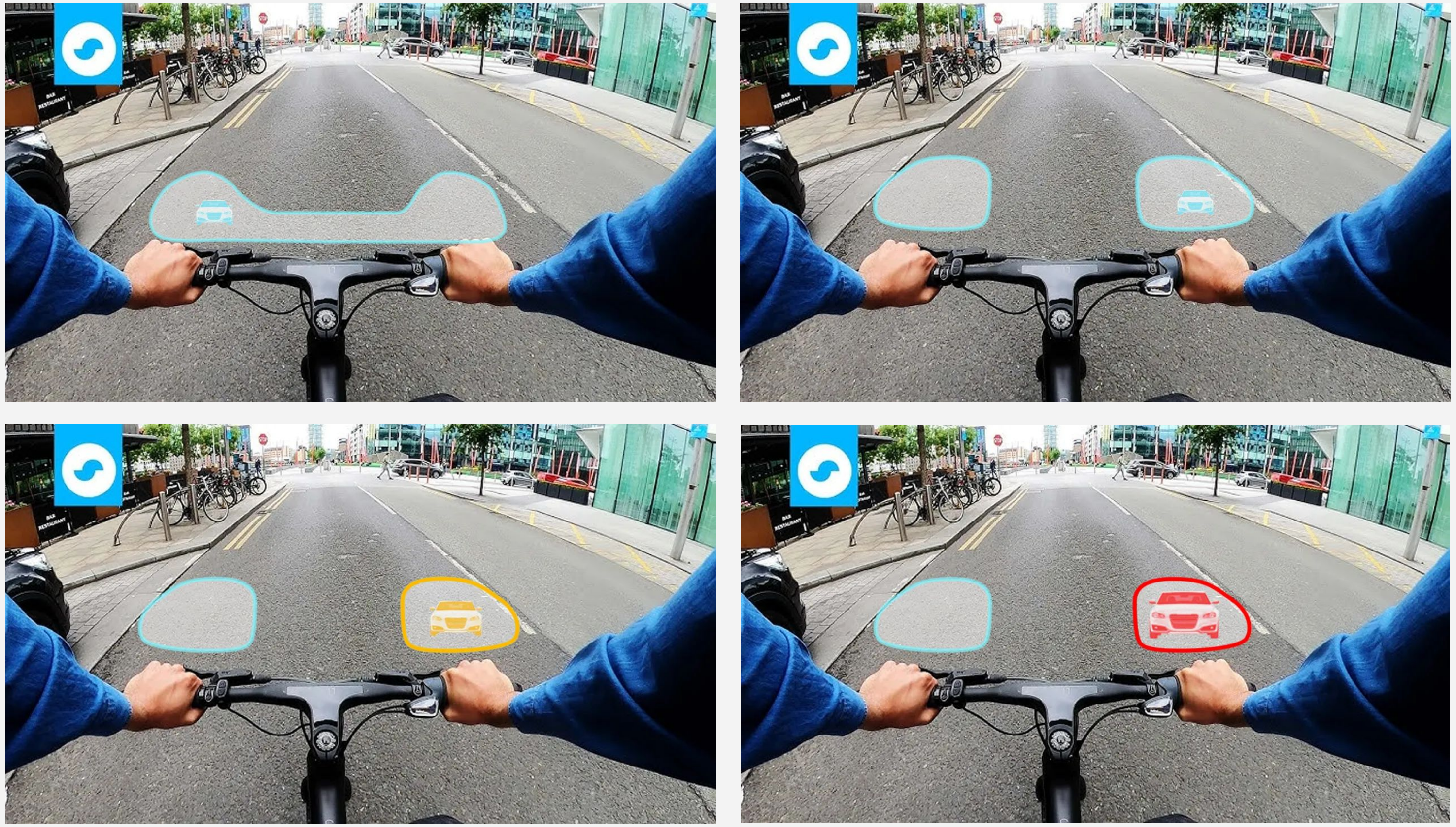
No 'Look-Back' Behaviour

No longer need to physically turn their heads, keeping forward focus and stability.

Design

Camera + Mirror-like Interface

Therefore, our project integrates camera and an familiar back-mirror AR interface to offer superior rear situational awareness and reduce cognitive burden.



Methodology and Approach

Research Questions

RQ1: How do different AR rearview mirror visualisation designs influence cyclists' perception of **rear hazards** in a simulated **virtual reality environment?**

RQ2: How does an AR rearview mirror-style interface influence cyclists' perception of **overtaking motorcycles** compared to simple directional rear hazard warnings?

System

- **AR**: Microsoft Hololens
- **Camera**: GoPro, BIKET, SEWEEK
- **App**: Unity

UI Design - Comparison Needed!

Fixed UI

Visual cues relative to handlebars, dynamically moving with perspective.
Color and size indicate proximity.

Anchored UI

Cues fixed on display.
semi-transparent by default, becoming opaque upon hazard detection.

Variance UI

alternative visual styling (shapes, patterns, animations).
No transparency adjustment needed.

Literature Review

1. Safety Sense: Enhancing Urban Cyclists' Perceived Safety during Motorcycle…

Our project directly addresses the limitations of "Safety Sense.", by enhancing existing UI and possibly sensing system.
(Huang, Y.-T. CHI 2025)

2. ProxiCycle: Passively Mapping Cyclist Safety Using Smart Handlebars for Near-Miss Detection

This paper collects 'close-pass' data between car and bicycle to **build a safety map**. It's a **macro-level** data for long-term route, whereas ours is a **micro-level** system for immediate hazard avoidance (Joseph Breda et al. CHI 2025)

3. The Impact of Augmented Reality-Based Visual Alerts on the Safety and Experience of E-bike riders

This study used a VR cycling simulator to test different AR visual alerts, giving us **useful test conditions** such as reaction time, perceived safety, and cognitive load (Junyao Fan et al. Cross Cultural Design HCII 2025)

4. Bicycle crash contributory factors: A systematic review

This paper addresses that biker's inattentiveness is one of a major crash causes, which validates our motivation.
(Salmon, P. M. Safety Science 2022)

5. Safety Performance Assessment via Virtual Simulation of V2X Warning Triggers…

The paper provides cyclist behavioral model for evaluating virtual safety simulation
(Lars Schories et al. Sustainability 2024)

Contribution

New AR Interface

Evaluation of an AR rearview **mirror interfaces** tailored for cyclists

Sensing System

Integration of **camera** with object detection/periodic display strategy

Cognitive Load

By eliminating the need for **'Look Back' head turns**, we anticipate cyclists to maintain focus on the road ahead.

Tentative Schedule

Week	Period	Activity
1	9/29 - 10/5	Finalise research questions and scope, basic system design
2	10/6 - 10/12	Finalise the detailed design and visualisation guidelines for the AR rearview mirror interface. (Size, color, hologram effects, information display methods, etc.)
3	10/13 - 10/19	Development of Sensor Integration and Data Processing Modules and internal testing of sensor reliability
4	10/20 - 10/26	AR Interface Development (Visualisation); Hazard Detection and Feedback System Integration; Refinement of the design based on early tests
5	10/27 - 11/2	Initial prototype testing and debugging; Iteration cycle 1: refine prototype
6	11/3 - 11/9	Experimental Protocol Design and incorporate feedback from prototype iteration
7	11/10 - 11/16	Experimental Environment Setup and Preliminary Testing; Iteration cycle 2: refine setup/interface
8	11/17 - 11/23	Experiment execution and data collection
9	11/24 - 11/30	Data analysis and derivation of results
10	12/1 - 12/8	Finalising the paper

Thank You !

Questions?